SMARTBRIDGE
Let's Bridge the Gap

Project Proposal

Project Initialization & Planning Phase

SKILLWALLET
A SmartBridge Product

Date	15 March 2024	**Team ID**	TEAM-FUEL-08
Project Title	Global Fuel Price & Taxation Trends	**Maximum Marks**	3 Marks

Project Overview & Proposed Solution

1. PROJECT OVERVIEW	
Objective	Build an interactive Business Intelligence dashboard on Tableau to benchmark global fuel pricing (USD/Liter), national taxation percentages, and historical Brent Crude oil price relationships.
Scope	Covers multi-country fuel dataset analysis across income brackets (High, Middle, Low), petrol vs diesel cost comparisons, LPG pricing metrics, and country-level tax rate evaluations (e.g. Germany vs Rwanda) sourced from the global fuel dataset.
2. PROBLEM STATEMENT	
Description	Global retail fuel prices (Petrol, Diesel, LPG in USD/Liter) fluctuate dramatically due to crude oil market volatility (Brent Crude) and disparate domestic tax policies, leaving policy analysts and fleet managers without a unified comparative analytical tool.
Impact	Enables transparent international taxation benchmarking, optimizes commercial cross-border transportation budgets, and supports evidence-based energy policy research.
3. PROPOSED SOLUTION	
Approach	Ingest global fuel (`global_fuel`) and taxation datasets, normalize prices to `petrol_usd_liter` and `diesel_usd_liter`, categorize by `income_level` and `subsidy_level`, compute regional KPI metrics, and construct interactive scatter plots, geospatial maps, and income-level bar charts.
Key Features	Dynamic country cross-filtering, Petrol vs Diesel margin comparison, `tax_percentage` vs fuel price scatterplot correlation, and `brent_crude_usd` historical trend visualizers.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources (CPU/GPU)	Quad-Core Intel i7 / 8-core CPU
	Memory	16 GB RAM
	Storage	512 GB SSD
Software	Frameworks / Tools	Tableau Desktop, Tableau Public
	Libraries	Pandas, NumPy, Matplotlib (Data Prep)
	Development Environment	Jupyter Notebook, VS Code, Git
Data	Global Fuel & Tax Repository Dataset	`global fuel/` Repository Extract (`petrol_usd_liter`, `diesel_usd_liter`, `lpg_usd_liter`, `brent_crude_usd`, `tax_percentage`, `income_level`)

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